

Vinh Ly

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Summary

Software Engineer with **18 years** of experience designing and shipping systems at scale across consumer products, infrastructure, and live-streaming platforms. Owned end-to-end reliability for **YouTube** playback infrastructure serving billions of daily requests, led cross-functional initiatives for **Google** Search indexing systems processing petabyte-scale datasets, and architected real-time features for **Whatnot** marketplace handling thousands of concurrent streams. Delivered measurable reductions in infrastructure costs and latency improvements across all tenures. Seeking to apply deep backend systems knowledge and technical leadership to high-impact product engineering challenges. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages: Python, C++, Java, SQL, Bash, TypeScript, JavaScript, HTML, CSS

Infrastructure: Distributed Systems, Google Cloud Platform, Kubernetes, Docker, Terraform, CI/CD

Data: Large-Scale Data Processing, Apache Beam, BigQuery, MapReduce, Data Pipelines

Platforms: Backend Systems, API Design, Microservices, On-call Operations, Performance Optimization

ML/AI: Machine Learning Infrastructure, Feature Engineering, Model Serving, A/B Testing, Node.js

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Architected playback infrastructure components supporting video delivery across multiple device platforms with focus on latency reduction and reliability improvements.
- Led technical discussions with partner teams to define API contracts and integration points for new video pipeline features.
- Designed backend services handling billions of daily requests with attention to horizontal scalability and fault tolerance requirements.
- Mentored engineers on systems design principles and code review practices to improved overall team technical quality.
- Evaluated infrastructure tradeoffs for cost optimization while maintaining service level objectives for core playback paths.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Reduced checkout failure rates by 40% through analysis of payment processing failures and implementation of retry mechanisms.
- Built real-time bidding and auction services supporting thousands of concurrent live streams with sub-second latency requirements.
- Led migration of legacy services to **Kubernetes**-based deployment infrastructure improving deployment frequency and system reliability.
- Established observability standards with structured logging and distributed tracing across marketplace backend systems.
- Designed event-driven architecture using **Python** microservices processing high-throughput marketplace transactions.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Owned on-call rotation for search infrastructure serving billions of queries daily with emphasis on incident response and root cause analysis.

- Led team of 6 engineers building indexing infrastructure for **Google Search** processing petabyte-scale data with **C++** and **MapReduce** pipelines.
- Delivered 25% reduction in indexing pipeline latency through redesign of document processing workflows and parallelization strategies.
- Collaborated with ML teams to deploy ranking model features through **BigQuery** feature stores serving production traffic.
- Established engineering standards and technical roadmap for search infrastructure modernization initiative spanning multiple quarters.

Software Engineer | Google | 02/2011 - 02/2015

- Built automated testing frameworks reducing integration bugs in ranking pipeline by 30% through comprehensive regression coverage.
- Optimized memory usage in core search serving processes resulting in 15% improvement in per-machine query throughput.
- Developed **C++** components for **Search** ranking infrastructure handling user-facing query traffic with strict latency SLAs.
- Contributed to design documents for next-generation indexing architecture adopted across multiple search product lines.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Authored automated test suites covering edge cases in platform API implementations reducing customer-reported bug rates.
- Participated in design reviews and code reviews for features shipping in Windows operating system releases.
- Developed **C++** systems components for Windows platform infrastructure with focus on kernel-level integration points.
- Collaborated with cross-functional teams including test, program management, and documentation for on-time feature delivery.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Completed intern project delivering **C++** utility tool adopted by team for automated build validation processes.
- Presented project findings to engineering organization demonstrating effective problem-solving and technical communication.

Education

B.S. Honors Degrees in Computer Science

Virginia Tech | 2004 - 2007